

FPS-3000 SBC — VersaBus address map

Every off-board address the firmware touches. All data accesses are 16-bit words except the mailbox, which is a 32-bit read. Derived from the disassembly, emulator bus logs and a post-boot RAM dump; see versabus_access_map.md.

Overview

Range	Card	Contents
\$FF0000-\$FF00FF	AP I/F	command register + four 32-byte channel windows
\$FF0200-\$FF021B	XLTR	mode / select / data / status / IRQ-mask file
\$FF0230-\$FF025F	XLTR	three MC68153-style BIMs (interrupt control + vectors)
\$70001C	chassis	mailbox, 32-bit read, bit 29 = host attention
\$400000-\$404000	chassis	address computations only; never read or written

AP I/F channel windows — base \$FF0040 + \$20-N

Offset	Dir	ch1	ch2	ch3	ch4	Role
+\$04	W	\$FF0044	\$FF0064	\$FF0084	\$FF00A4	write port
+\$08	R	\$FF0048	\$FF0068	\$FF0088	\$FF00A8	read A: this read consumes a host byte
+\$0A	R	\$FF004A	\$FF006A	\$FF008A	\$FF00AA	status; host presents \$4F
+\$0E	R	\$FF004E	\$FF006E	\$FF008E	\$FF00AE	read B

\$FF000E = command / argument register, write-only, 8 write sites. A plain boot reads only the +\$0E port of each channel, twice each. \$FF0048 is never read, so a queued host byte is never consumed.

XLTR control file

Addr	Name	R	W	Observed in a boot	Values seen
\$FF0200	MODE0	20	20	R1 / W1	0
\$FF0202	MODE1	37	32	R2 / W6	8020 8000 8021 9021
\$FF0204	CHANNEL_SELECT	26	80	R0 / W1	281
\$FF020C	COUNTER	1	9	never executed	
\$FF0210	MODE2	3	15	R1 / W5	F 0
\$FF0214	DATA_LO	0	1	never executed	
\$FF0216	DATA_HI	3	20	R0 / W2	C0
\$FF0218	STATUS_IRQ	22	44	never executed	
\$FF021A	IRQ_MASK	51	51	never executed	

R and W count instruction sites in the disassembly. The four never-executed registers are the panel-command path, which waits on a chassis response we do not yet generate.

The three BIMs — CR0-3 at +\$0/+\$2/+\$4/+\$6, VR0-3 at +\$8/+\$A/+\$C/+\$E

BIM	ch	CR	value	level	VR	vec	vec addr	handler	owner
0	0	\$FF0230	\$5E	6	\$FF0238	\$41	\$104	F04930	TCBRDHC
0	1	\$FF0232	\$00	off	\$FF023A	\$42	\$108	F00896	generic
0	2	\$FF0234	\$00	off	\$FF023C	\$43	\$10C	F00896	generic
0	3	\$FF0236	\$00	off	\$FF023E	\$44	\$110	F00896	generic
1	0	\$FF0240	--	--	\$FF0248	--	--	--	unused
1	1	\$FF0242	\$00	off	\$FF024A	\$49	\$124	F0A27A	panic (spare)
1	2	\$FF0244	\$5F	7	\$FF024C	\$45	\$114	F07EE6	TCBXP1I
1	3	\$FF0246	\$5F	7	\$FF024E	\$46	\$118	F074E6	TCBXP2I
2	0	\$FF0250	\$5F	7	\$FF0258	\$47	\$11C	F06AE6	TCBXP3I
2	1	\$FF0252	\$5F	7	\$FF025A	\$48	\$120	F060CE	TCBXP4I
2	2	\$FF0254	\$5F	7	\$FF025C	\$4A	\$128	F05DD6	TCBI01I
2	3	\$FF0256	\$00	off	\$FF025E	--	--	--	disabled

CR bits: 0-2 = IRQ request level (0 disables the channel), 4 = IRE (enable), 5 = IRAC (auto-clear, off here), 7 = Flag. The BIM puts its vector register on the data bus during IACK, so these interrupts are vectored rather than autovectored. Vector number x 4 = vector address.

Vector table after boot

Vector	Addr	Handler	Meaning
\$41-\$44	\$104-\$110	F04930 / F00896 x3	BIM0: TCBRDHC, then three unused
\$45-\$48	\$114-\$120	F07EE6 F074E6 F06AE6 F060CE	XP-32 channels 1-4
\$49	\$124	F0A27A	panic catch-all (spare channel)
\$4A	\$128	F05DD6	TCBI01I host link
autovec L4	\$070	F00EC8	MC6840 PTM tick: the only live autovector
autovec L7	\$07C	00000000	never installed; level 7 is vectored here

VersaBus P1 pinout — the signals these accesses ride on

From refs_extracted/versabus_pinout.md, extracted from the Motorola VERSAbus Specification (M68KVBS) and the M68KVM02 manual. All strobes active low.

Group	Pins	Signals
Data D00-D15	5-20	5=D00 6=D01 7=D02 8=D03 9=D04 10=D05 11=D06 12=D07
		13=D08 14=D09 15=D10 16=D11 17=D12 18=D13 19=D14 20=D15
Address A01-A23	36-58	36=A01 37=A02 38=A03 39=A04 40=A05 41=A06 ... 58=A23
	35	LWORD
Transfer control	30 / 34	AS (address strobe) / WRITE (high=read, low=write)
	25 / 27	DS0 (low byte D0-D7) / DTACK (slave handshake)
Errors	81 / 80	BERR (bus error) / SYSFAIL
System	74 / 78	SYSRESET / ACFAIL

Interrupts (open collector, wired-OR on the backplane; the CPU resolves priority on IPL2-IPL0):

IRQ1	IRQ2	IRQ3	IRQ4	IRQ5	IRQ6	IRQ7
87	88	89	90	91	92	93

The BIM asserts one of these according to bits 0-2 of its control register. \$5F selects level 7, so the five enabled channels should drive P1 pin 93; the \$5E channel should drive pin 92. Neither has been verified on the board.

P2 serial (the monitor's link): 73 = TXD1 (SBC out), 75 = RXD1 (SBC in), 1-6 = GND. The RS-232 drivers need +/-12V, which the chassis does not supply.

Open

The IACKIN / IACKOUT daisy-chain order between the three BIMs is untraced, so arbitration between the five level-7 channels is unknown. Whether \$5F really reaches IRQ7 on this board is a continuity check away: BIM pin 17 to VersaBus P1 pin 93. See versabus_trace_worksheet.pdf.